Training Parts vs. Matching Parts: A Controlled Study of Part-Level Objectives for Occlusion-Robust Pose Gait Recognition

Minas Aslanyan[1] and Levon Aslanyan[1]

Institute for Informatics and Automation Problems, National Academy of Sciences of the Republic of Armenia (IIAP), Yerevan, Armenia
minas.aslanyan@iiap.sci.am, lasl@sci.am

Abstract. Part-level learning is a plausible response to occluded gait: if one limb is missing, the remaining anatomical regions should still support retrieval. However, it is unclear whether parts should be supervised during training, used only during matching, or reliability-gated in both stages. We conduct a controlled development and official-test study on 2D pose sequences from CASIA-B. On a 12-identity development split, a common multistream graph backbone is trained under six objectives using three seeds. The metric baseline is strongest: adding part supervised contrastive learning reduces averaged corrupted-probe Rank-1 from 63.90% to 57.78%, a paired difference of -6.12 percentage points with a hierarchical 95% bootstrap interval of $[-9.54, -2.68]$. Freezing the baseline checkpoints and replacing the untrained local projections with normalized pre-projection part pools raises the same endpoint to 68.23%, a gain of $+4.32$ points, $[+3.25, +5.35]$. We then retrain the metric baseline on all 74 official training identities with five seeds and evaluate the untouched 50-identity test split. Under the standard single-view CASIA-B protocol, global matching obtains $48.35 \pm 1.85\%$ average Rank-1, whereas reliability-weighted raw-part matching obtains $54.04 \pm 2.18\%$. The paired gain is $+5.69$ points with a 95% interval of $[+5.19, +6.20]$. Pooled multi-view test outcomes and synthetic corruptions also favor raw-part matching, with gains of $+6.58$, $+6.39$, and $+5.63$ points for uncorrupted, corrupted-probe, and independently corrupted-gallery endpoints. Uniform raw-part averaging reaches the same 54.04% standard-protocol average, so reliability gating is not a material independent contribution. The evidence rejects the tested part-level training objectives while supporting a simpler conclusion: match anatomical features already present in the globally trained backbone without learning a local projection. Real-occlusion and external- dataset confirmation remain necessary.

Keywords: gait recognition · 2D pose · occlusion robustness · supervised contrastive learning · part-based matching · negative results

1 Introduction

Gait recognition supports identity retrieval at a distance, including cases in which face evidence is weak or unavailable. Pose-based methods replace raw appearance with joint trajectories and can reduce dependence on clothing, background, and illumination [12]. Graph encoders such as ST-GCN [4], GaitGraph [6], and GaitGraph2 [13] make this representation computationally natural. Its main weakness is equally direct: a pose sequence contains only the body evidence recovered by the pose estimator. Occlusion, truncation, detector uncertainty, and identity overlap can remove joints or produce anatomically inconsistent tracks. Recent surveys and dedicated occluded-gait benchmarks consequently identify partial body evidence as a central challenge [8,9].

An intuitive remedy is to learn anatomical embeddings. Supervised contrastive learning can align two corrupted observations of the same identity while separating different identities [3]. A part-level extension can apply the same objective independently to the torso and four limbs, weighting each term by the observed pose confidence. This argument is plausible, but it combines three decisions that need not have the same answer: whether to train local heads, whether to gate their training loss, and whether to use local scores during retrieval. A method can benefit from part-aware matching even when an auxiliary part loss conflicts with the primary identity objective. Conversely, a trained local head can collapse or overfit synthetic missingness.

This paper separates these decisions in a controlled factorial development study followed by a fixed official-test confirmation. Every configuration uses the same 2D input, augmentation distribution, multistream graph backbone, batch construction, clean classification anchor, global batch-hard triplet loss, checkpoint-selection rule, and evaluation masks. Only the auxiliary global or part metric term changes. Each configuration is trained with seeds 11, 22, and 33 on the development split. After selecting the raw-part matcher, we retrain only the metric baseline on all 74 official training identities using seeds 11, 22, 33, 44, and 55 and evaluate identities 075–124. Saved probe-level correctness outcomes permit paired resampling over identities rather than treating aggregate accuracy cells as independent observations.

The study asks five questions:

1. Does global supervised contrastive learning improve a strong classification-plus-triplet baseline?
2. Does part-level supervised contrastive learning improve robustness, and does reliability gating explain its effect?
3. Does a part-level triplet objective provide a better-trained local descriptor than part supervised contrastive learning?
4. Is part-aware matching useful independently of part-aware training?
5. Does the selected matcher carry over to the official CASIA-B test identities without further tuning?

The observed answers are negative for auxiliary part training and positive for local matching. Global SupCon is statistically inconclusive relative to the baseline. Part SupCon produces a clear robustness loss. Ungating recovers only part

of the deficit. Part triplet collapses its local embedding variance and does not improve retrieval. In contrast, deterministic matching of normalized pre-projection part pools improves clean and corrupted recognition for every training seed and all four corruption families. Ten freshly initialized frozen heads confirm that the earlier gain is not dependent on one favorable random projection; nevertheless, the raw descriptor is stronger and removes the projection entirely. The official-test experiment preserves this conclusion: both raw-part variants improve NM, BG, and CL recognition over global matching, while reliability-weighted and uniform part averaging remain effectively tied.

The contribution is therefore not a state-of-the-art accuracy claim. It is an auditable diagnosis of a common design assumption, supported by a complete 6×3 run matrix, checkpoint-only controls spanning ten random projections and five fusion weights, paired probe outcomes, and training diagnostics. A separate five-seed official-test experiment confirms the selected parameter-free descriptor under the standard CASIA-B protocol and controlled synthetic corruptions. The results illustrate why clean accuracy alone is insufficient for selecting an occlusion-robust gait objective.

2 Related Work

2.1 Contrastive learning for structured motion

SimCLR and MoCo established the importance of positive-pair construction, projection heads, and negative sampling in visual contrastive learning [1,2]. Supervised contrastive learning extends the positive set to all same-class observations [3]. For skeleton sequences, domain-specific transformations are required because an arbitrary image augmentation need not preserve joint semantics. Skeleton- Contrastive learning uses multiple skeleton representations and motion-aware transformations for action recognition [5]; prototype contrastive learning has also been applied to skeleton person re-identification [11]. More recently, GaitPT studied model, data, and compute scaling for self-supervised skeleton gait pretraining on millions of in-the-wild sequences [25]. These results motivate contrastive objectives, but they do not establish that independent anatomical objectives help fine-grained gait identity retrieval.

2.2 Pose-based gait and occlusion

PoseGait established a compact 3D-pose route to model-based recognition [12]. GaitGraph and GaitGraph2 subsequently demonstrated graph convolution with learned spatial–temporal and multistream skeleton features [6,13]. GPGait adds human-oriented normalization and a part-aware graph network for cross-dataset generalization [14], while SkeletonGait converts joints to heatmap-like skeleton maps to recover structural cues that coordinate-only models may underuse [15]. OpenGait emphasizes standardized protocols and reproducible evaluation across gait datasets [7].

Occlusion is heterogeneous: random joint loss, a missing limb, a continuous temporal interruption, and a real detector failure produce different evidence distributions [8]. OccGait and GaitMoE further show that real occlusion combines missing information with misalignment and noise [9]. Related skeleton work has combined partial spatio-temporal learning with imputation in occluded action recognition [10]. Large in-the-wild benchmarks such as GREW and Gait3D also expose shifts and nuisance factors absent from controlled laboratory data [23,24]. Our study does not claim that synthetic masks substitute for these real errors. Instead, it uses deterministic synthetic families to isolate objective and matching effects before external validation.

2.3 Part-aware gait representation

Part modeling is well established in appearance-based gait recognition. GaitPart assigns separate short-range temporal models to horizontal body regions [16]; GaitGL couples global and local feature extraction [17]; and 3D Local CNNs learn part-specific spatio-temporal neighborhoods [18]. GPGait is more directly related to our pose setting because its graph partition explicitly captures local–global keypoint relations [14]. These works support the value of anatomical structure, but they jointly change representation, architecture, and training. They therefore do not answer the narrower question studied here: with a fixed pose backbone, should local parts receive their own metric objective, enter the matcher, or both?

2.4 Position of this study

Most method papers compare a complete proposal with a small set of baselines. Here the main object is the interaction between training objective and retrieval descriptor. All methods share the same backbone and occlusion exposure. We explicitly retain results that contradict the original hypothesis and use run-level artifacts to distinguish three possible explanations: ineffective global contrast, harmful local supervision, and descriptor-head behavior.

3 Controlled Study

Figure 1 summarizes the controlled design. Training changes only the auxiliary objective attached to shared global or anatomical features; the checkpoint-only control study freezes the baseline encoder and changes only the local descriptor and matching rule.

3.1 Data and identity-disjoint evaluation

We use the public HRNet-derived [20] 2D pose release for CASIA-B [19]. Each frame contains 17 joints following the COCO keypoint convention [21], with two coordinates and a continuous confidence value. For development, a fixed split

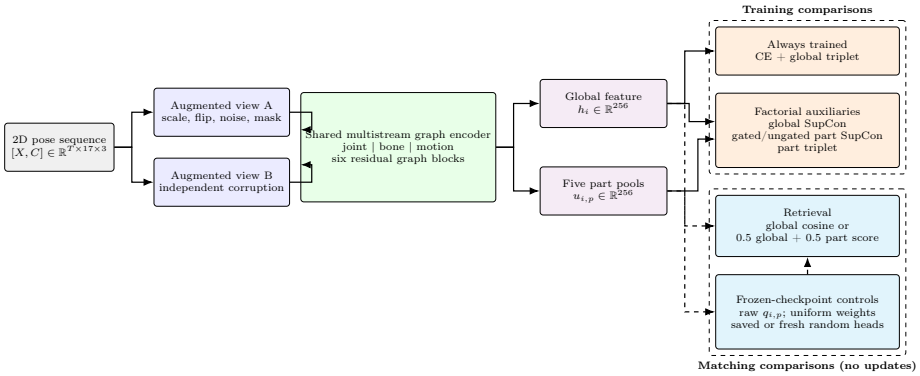


Fig. 1. Separation of part-aware training from part-aware matching. Solid arrows denote the shared forward path. The upper branch contains the six training objectives; the lower branch contains frozen-checkpoint descriptor controls and performs no parameter updates.

seed (2026) holds out identities {002, 006, 012, 025, 026, 032, 043, 046, 054, 059, 061, 072} for validation; the remaining 62 identities train the models. For every held-out identity, normal sequences 01–04 form the gallery and normal sequences 05–06, bag sequences 01–02, and clothing sequences 01–02 form the probe set. This gives 792 probe sequences. Identical-view gallery entries are excluded when a probe is scored.

The official confirmation is disjoint from this development stage. It trains the selected metric baseline on identities 001–074 and evaluates identities 075–124. The standard single-view protocol uses normal sequences 01–04 as gallery and normal 05–06, bag 01–02, and clothing 01–02 as probes, averages over the 11 gallery views, and excludes identical-view matches. The resulting test set contains 3,300 probe sequences from 50 identities. We additionally use a pooled multi-view gallery for the controlled synthetic-corruption endpoints; this pooled protocol is reported separately and is not presented as the standard single-view score.

Coordinates are centered on the available pelvis joints, falling back to the visible-joint centroid if the pelvis is missing, and are normalized by sequence body scale. Confidence is supplied as a separate input and pooling weight; it does not multiply and geometrically shrink visible coordinates. Training uses contiguous 60-frame clips, with cyclic repetition for shorter sequences. Evaluation uses every real frame in each sequence.

3.2 Multistream graph backbone

Let $X \in \mathbb{R}^{T\times 17\times 2}$ be a normalized pose and $C \in [0,1]^{T\times 17}$ its confidence. The encoder constructs three input streams: joints $[X, C]$; bones, defined relative to a rooted skeleton parent and observed only when both endpoints are available; and

motion at lags one and two, again retaining confidence only when both temporal endpoints are observed. Stream-specific stems feed six residual graph blocks. Each block uses learned perturbations of the physical adjacency and temporal convolutions with dilations 1, 2, and 4. The hidden width grows from 64 to 256, and confidence- weighted global pooling produces a 256-dimensional normalized embedding h_i.

For local representations, we partition the COCO skeleton into torso, left and right arms, and left and right legs. Confidence-weighted pooling within part p produces $u_{i,p} \in \mathbb{R}^{256}$. The deterministic raw descriptor is $q_{i,p} = u_{i,p}/\|u_{i,p}\|_2$. The original learned alternative passes $u_{i,p}$ through a part-specific two-layer projection head and ℓ_2 normalization to obtain $z_{i,p}$. Part reliability is

$$r_{i,p} = \frac{1}{T_i|p|} \sum_{t=1}^{T_i} \sum_{j \in p} C_{i,t,j}, \tag{1}$$

where T_i counts real rather than padded frames.

3.3 Shared training recipe

Each batch samples 16 identities and four sequences per identity. Two views of each sequence receive scale, horizontal-flip, and coordinate-noise augmentation. Each view independently has 0.3 probability of remaining geometrically complete; otherwise one of random-joint, body-part, or temporal masking is sampled. The maximum training severity increases linearly from 0.05 to 0.5 over the first half of training. Dynamic-joint corruption is held out from the training sampler and used only for evaluation.

All configurations share the base objective

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{CE,aug}} + 0.5\mathcal{L}_{\text{CE,clean}} + \mathcal{L}_{\text{Tri}}, \tag{2}$$

where the first term classifies both augmented views, the clean anchor classifies the original sequence, and $\mathcal{L}_{\text{Tri}}$ is batch-hard triplet loss [22] on h_i. Same-source augmented views are excluded as triplet positives, so positives must be distinct sequences of the same identity.

Global supervised contrastive loss $\mathcal{L}_{\text{gSC}}$ operates on a projection of the global embeddings and uses every same-identity view as a positive. Part supervised contrastive loss applies the same construction independently to each $z_{i,p}$. In the gated version, a positive pair receives weight $r_{i,p}r_{j,p}$ and the anchor receives weight $r_{i,p}$; parts below reliability 0.05 are excluded from both positives and negatives. The ungated control gives all present parts unit weight. Part triplet loss $\mathcal{L}_{\text{pTri}}$ applies batch-hard triplet independently to every present part. The complete factorial objective is

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \gamma_g\mathcal{L}_{\text{gSC}} + \gamma_p\mathcal{L}_{\text{pSC}} + \gamma_{pt}\mathcal{L}_{\text{pTri}}. \tag{3}$$

Development training lasts 160 epochs with AdamW, initial learning rate 10^{-3}, weight decay 10^{-4}, five warm-up epochs, cosine decay to 1% of the initial

Table 1. Controlled training objectives. All unlisted terms and all backbone, augmentation, and optimization settings are identical.

Configuration	γ_g	γ_p	γ_{pt}	Part-SupCon gate
Metric baseline	0	0	0	–
Global SupCon	0.2	0	0	–
Part SupCon	0	0.2	0	reliability
Part triplet	0	0	0.2	–
Global + part SupCon	0.2	0.2	0	reliability
Global + part SupCon (ungated)	0.2	0.2	0	presence only

rate, label smoothing 0.1, global SupCon temperature 0.07, part temperature 0.1, triplet margin 0.2, gradient clipping at 5, and automatic mixed precision. A checkpoint is selected every five epochs using only the held-out validation split. The score assigns 75% weight to clean recognition and 25% to the mean of body-part and random-joint recognition at severity 0.3. Selection uses only the global descriptor for every method.

For official confirmation, only the metric baseline is retrained. All 74 official training identities are used for a fixed 85 epochs with the same base objective and optimization settings. Validation is disabled, no official test outcome is used for checkpoint selection, and epoch 85 is evaluated for each of five seeds (11, 22, 33, 44, and 55). The global, uniform raw-part, and reliability-weighted raw-part descriptors are computed from each same-seed checkpoint at the fixed 0.5 local-score weight.

3.4 Retrieval descriptors

The global descriptor uses cosine similarity, $s_g(i,j) = h_i^\top h_j$. The reliability-weighted part descriptor keeps the same global term and averages local cosine scores over jointly reliable parts. Let $v_{i,p}$ denote the local descriptor under evaluation: either the learned projection $z_{i,p}$ or the normalized raw pool $q_{i,p}$. Then

$$s_p(i,j) = \frac{\sum_p r_{i,p} r_{j,p} v_{i,p}^\top v_{j,p}}{\max(\sum_p r_{i,p} r_{j,p}, 10^{-8})}, \tag{4}$$

$$s_{\mathrm{rel}}(i,j) = 0.5 s_g(i,j) + 0.5 s_p(i,j). \tag{5}$$

If no local part is jointly available, matching falls back to s_g. Both descriptors are evaluated for every checkpoint. Importantly, the metric baseline and global-SupCon configurations give the part heads no direct loss; their part projection weights therefore remain at their seeded initialization, although the shared backbone changes during training.

We resolve this confound in a checkpoint-only control study. The three selected metric-baseline backbones are frozen and each sequence is encoded once

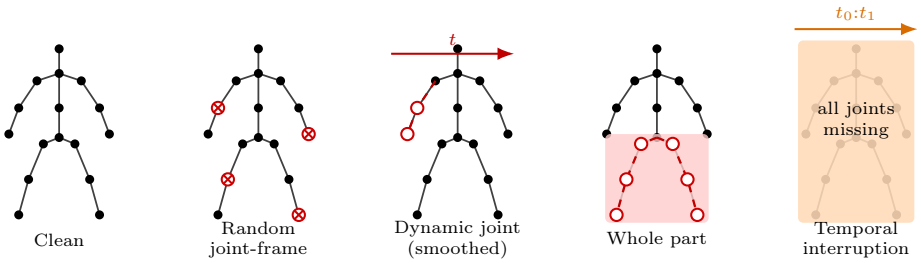


Fig. 2. Conceptual corruption families. Red marks unavailable joints or body regions; orange marks a contiguous interval in which the complete pose is unavailable. Random joint-frame masks vary independently, whereas dynamic-joint masks are temporally smoothed.

per corruption realization. We compare (i) $q_{i,p}$ with reliability-weighted matching; (ii) $q_{i,p}$ with uniform weights over jointly present parts; (iii) the saved untrained heads; and (iv) ten freshly initialized frozen copies of the same nonlinear head architecture per backbone checkpoint. No condition updates model parameters. The local-score weight is fixed at 0.5 for the primary comparison. We additionally report weights $\{0, 0.25, 0.5, 0.75, 1\}$ as sensitivity analysis; they are not used to replace the pre-specified primary weight after observing validation performance.

3.5 Occlusion protocols and outcomes

We evaluate random joint-frame removal, temporally smoothed dynamic-joint removal, whole-part removal, and a contiguous temporal interruption at severities $\rho \in \{0, 0.1, 0.3, 0.5, 0.7\}$. Five deterministic corruption realizations are averaged at every positive severity. In the clean-gallery protocol only probes are corrupted. In the occluded-gallery protocol gallery and probe sequences receive independently seeded corruption of the same family and severity. We additionally report the CASIA-B-style single-view-gallery average over NM, BG, and CL. Development tables use held-out training identities; the dedicated official-test table uses identities 075–124 and is labeled explicitly.

Figure 2 illustrates the four corruption families, and Figure 3 records where each family appears in training and in the two gallery protocols.

The primary measure is Rank-1 retrieval. Rank-5, Rank-10, mean average precision, equal-error rate, and true-acceptance rates are retained in the released run artifacts but are secondary to the present diagnosis. Development tables report the mean and sample standard deviation over three training seeds. For paired development comparisons, each saved outcome is a per-probe correctness value, averaged over corruption realizations where applicable. We first average the selected corruption cells within each probe, compute candidate-minus-baseline differences within each of 12 identities and three matched seeds, and then resample seeds and identities hierarchically with replacement for 20,000

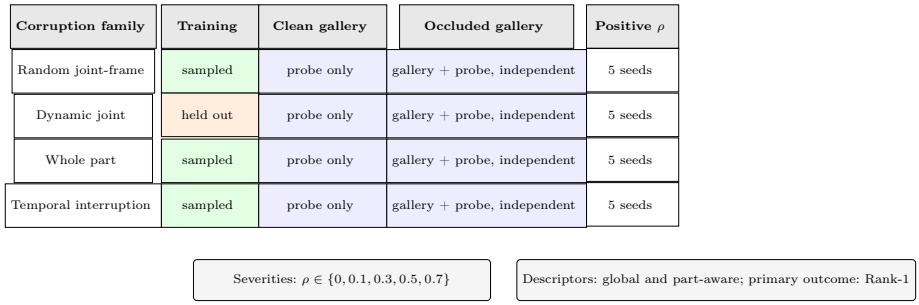


Fig. 3. Evaluation matrix. Dynamic-joint corruption is deliberately held out from training. Positive-severity results average five deterministic corruption realizations, and gallery and probe corruptions are independently seeded in the occluded-gallery protocol.

draws. Intervals are percentile 95% intervals. They are used to express uncertainty, not to support a confirmatory significance claim across the many exploratory endpoints.

Official-test tables report mean and sample standard deviation over five matched training seeds. For the standard single-view aggregate, only seed-level paired outcomes are retained, so candidate-minus-global uncertainty is a paired t interval over the five seed differences. For pooled multi-view and synthetic-corruption endpoints, saved per-probe outcomes support a crossed bootstrap: five seeds and 50 test identities are independently resampled with replacement for 100,000 draws while preserving candidate–baseline pairing. Synthetic endpoints first average the four positive severities and four corruption families within each probe.

4 Results

4.1 Global contrast is neutral; part contrast is harmful

Table 2 shows the complete factorial. With global matching, the metric baseline attains 81.52% clean Rank-1, 63.90% averaged positive-severity clean-gallery Rank-1, and 67.93% when both gallery and probe are corrupted. Global SupCon produces 79.63%, 63.23%, and 67.42%, respectively. The paired intervals in Table 3 include zero for all four global-descriptor endpoints, so this experiment neither demonstrates a benefit nor a clear robustness penalty from the global SupCon term.

Part SupCon behaves differently. Its global descriptor loses 6.12 percentage points on the positive-severity clean-gallery aggregate, with interval $[-9.54, -2.68]$, and 8.68 points at severity 0.5, with interval $[-14.50, -3.49]$. The reliable-parts descriptor loses still more. Thus the failure is not limited to a poorly chosen local matching weight: the shared global representation also deteriorates when the local contrastive objective is added.

Table 2. CASIA-B validation Rank-1 (%). Clean is mean ± seed standard deviation. "All > 0" averages four corruption families at severities 0.1, 0.3, 0.5, and 0.7. Occluded gallery averages the same positive-severity grid. Single-view averages NM, BG, and CL. Every cell uses three training seeds.

Global descriptor					
Training objective	Clean	Clean gallery all > 0	Clean gallery $\rho = 0.5$	Occluded gallery all > 0	Single-view
Metric baseline	81.52 ± 2.43	63.90	64.25	67.93	67.36
Global SupCon	79.63 ± 3.23	63.23	64.15	67.42	66.18
Part SupCon	78.79 ± 1.14	57.78	55.57	64.26	65.33
Part triplet	81.27 ± 3.41	60.65	59.10	66.71	67.82
Global + part SupCon	80.05 ± 1.34	60.70	59.06	65.56	65.75
Global + part SupCon (ungated)	80.89 ± 1.59	61.72	60.59	66.99	66.29

Reliable-parts descriptor					
Training objective	Clean	Clean gallery all > 0	Clean gallery $\rho = 0.5$	Occluded gallery all > 0	Single-view
Metric baseline	83.04 ± 1.31	66.31	67.97	70.57	69.34
Global SupCon	82.79 ± 3.36	66.21	67.85	70.38	68.66
Part SupCon	75.42 ± 2.25	54.66	52.65	61.62	61.81
Part triplet	81.31 ± 3.42	60.66	59.11	66.73	67.82
Global + part SupCon	77.86 ± 4.45	57.48	54.84	63.18	63.40
Global + part SupCon (ungated)	78.20 ± 0.07	59.35	58.11	64.85	65.60

Table 3. Paired Rank-1 difference from the metric baseline in percentage points (hierarchical 95% bootstrap interval over seed and validation identity). Negative values favor the baseline.

Training objective	Descriptor	Clean	Clean gallery, all > 0
Global SupCon	Global	-1.89 [-4.59, +0.46]	-0.67 [-4.21, +3.35]
Part SupCon	Global	-2.74 [-6.02, +0.51]	-6.12 [-9.54, -2.68]
Part triplet	Global	-0.25 [-4.42, +3.54]	-3.26 [-6.57, +0.21]
Global + part SupCon	Global	-1.47 [-4.76, +1.98]	-3.21 [-8.08, +2.40]
Global + part (ungated)	Global	-0.63 [-4.63, +4.42]	-2.19 [-4.68, +0.41]
Global SupCon	Reliable-parts	-0.25 [-3.11, +2.36]	-0.10 [-2.60, +2.61]
Part SupCon	Reliable-parts	-7.62 [-10.86, -4.00]	-11.65 [-15.58, -8.03]
Part triplet	Reliable-parts	-1.73 [-5.94, +1.89]	-5.65 [-8.92, -2.17]
Global + part SupCon	Reliable-parts	-5.18 [-9.34, -0.63]	-8.83 [-13.78, -2.67]
Global + part (ungated)	Reliable-parts	-4.84 [-7.95, -1.81]	-6.96 [-10.56, -3.72]

Training objective	Descriptor	Clean gallery, $\rho = 0.5$	Occluded gallery, all > 0
Global SupCon	Global	-0.10 [-5.43, +6.03]	-0.51 [-2.58, +1.64]
Part SupCon	Global	-8.68 [-14.50, -3.49]	-3.67 [-5.70, -1.65]
Part triplet	Global	-5.15 [-9.71, -0.32]	-1.22 [-3.69, +0.99]
Global + part SupCon	Global	-5.19 [-12.57, +3.73]	-2.37 [-5.48, +1.02]
Global + part (ungated)	Global	-3.65 [-8.04, -0.10]	-0.94 [-2.57, +0.80]
Global SupCon	Reliable-parts	-0.12 [-3.46, +3.50]	-0.20 [-1.89, +1.59]
Part SupCon	Reliable-parts	-15.32 [-21.20, -10.15]	-8.96 [-11.32, -6.53]
Part triplet	Reliable-parts	-8.86 [-14.48, -3.24]	-3.85 [-5.97, -1.92]
Global + part SupCon	Reliable-parts	-13.13 [-19.81, -4.69]	-7.39 [-10.83, -3.42]
Global + part (ungated)	Reliable-parts	-9.85 [-15.36, -5.26]	-5.73 [-7.61, -3.91]

4.2 Reliability gating is not the primary cause

The combined gated objective reaches 60.70% on the positive-severity global aggregate. Removing reliability weights increases this to 61.72%, and raises the corresponding reliable-parts result from 57.48% to 59.35%. The recovery is directionally consistent but remains below the 63.90% and 66.31% metric baselines. Average part reliability (0.502) and presence rate (0.860) are essentially identical across methods because they are determined by the shared input sampler. Gating therefore changes how the same observed evidence weights the auxiliary loss; it does not explain the entire deficit.

4.3 Training a part head does not make the local matcher stronger

The part-triplet control closely preserves clean global accuracy (81.27%) but falls to 60.65% on the positive-severity aggregate and 59.10% at severity 0.5. Its reliable-parts and global results are nearly identical. The training diagnostics in Table 4 explain this behavior: final normalized part variance is approximately

Table 4. Training diagnostics averaged over three seeds. Part variance is the mean coordinate-wise variance over the final 20 epochs. Cosines are measured every five epochs; parentheses give nonzero finite cosine measurements out of 96 scheduled.

Objective	Variance ($\times 10^{-3}$)	Part–global	Part–triplet
Metric baseline	1.269	–	–
Global SupCon	1.300	–	–
Part SupCon	3.338	–	0.212 (95)
Part triplet	0.002	–	0.012 (9)
Global + part SupCon	3.349	0.265 (96)	0.203 (96)
Global + part SupCon (ungated)	3.242	0.241 (96)	0.195 (96)

Table 5. Checkpoint-only descriptor controls on CASIA-B validation Rank-1 (%). All conditions use the same three selected metric-baseline backbones and the fixed 0.5 fusion weight. "Fresh random heads" is the mean across ten projection seeds; brackets give the range after averaging the three backbone seeds. No control retrains the backbone or local head.

Descriptor	Clean	Clean gallery all > 0	Clean gallery $\rho = 0.5$	Occluded gallery all > 0
Global only	81.52	63.90	64.25	67.93
Saved frozen head	83.04	66.31	67.97	70.57
Fresh random heads	83.75 [83.21, 84.26]	66.50 [66.32, 66.73]	68.07 [67.71, 68.40]	70.70 [70.58, 70.81]
Raw parts, uniform	84.43	68.11	69.99	71.71
Raw parts, reliability weighted	**84.47**	**68.23**	**70.31**	**71.76**

1.7×10^{-6}, and a nonzero finite part–triplet cosine is available in only 9 of 96 scheduled diagnostics. The other diagnostics have a zero-norm gradient for at least one compared loss; they are not numerical failures. The trained local heads have effectively collapsed for this objective.

Part SupCon avoids variance collapse: its final variance is 3.34×10^{-3}. Nevertheless, its gradient is less aligned with the global triplet objective than the global SupCon gradient is. In the combined model, the mean part–triplet cosine is 0.203, compared with a global–triplet cosine of 0.446. These cosines are positive, so the data do not support a claim of direct gradient opposition. They instead show weaker agreement and leave open other causes, including excessive local invariance, part-specific identity ambiguity, or mismatch between the local contrastive task and retrieval.

4.4 Raw part matching resolves the random-head confound

Table 5 removes the learned projection from the same checkpoints. Reliability-weighted raw parts improve clean Rank-1 by 2.95 points over global matching, with hierarchical paired interval $[1.47, 4.67]$, and improve the positive-severity aggregate by 4.32 points, $[3.25, 5.35]$. The gain grows to 6.06 points at severity 0.5, $[4.12, 8.29]$, and remains 3.83 points with independently corrupted galleries, $[3.07, 4.72]$. Every training seed improves on both clean and corrupted-probe endpoints, and the four family-specific corrupted aggregates all increase.

All ten fresh random-head seeds also pass the clean and robustness thresholds: their clean gains range from 1.68 to 2.74 points and their corrupted-probe gains from 2.42 to 2.82 points. Thus the original improvement did not depend on one favorable initialization. However, removing the nonlinear projection is both deterministic and stronger. The effect is also insensitive to moderate fusion: raw reliable parts improve the corrupted aggregate by 2.30, 4.32, and 5.68 points at weights 0.25, 0.5, and 0.75, respectively. We retain 0.5 as the declared primary weight rather than selecting 0.75 after observing validation.

Reliability weighting is not the main source of the raw-descriptor gain. Relative to uniform averaging, it changes clean Rank-1 by only +0.04 points and the principal corrupted aggregate by +0.12 points. The latter interval, $[+0.02, +0.22]$, is positive but practically small and arises mainly under whole-part corruption. The supported mechanism is therefore anatomical pooling and score-level fusion, not a large general advantage from confidence gating.

4.5 Official CASIA-B test confirmation

Table 6. Five-seed CASIA-B official-test Rank-1 (%), reported as mean ± sample standard deviation. The upper panel follows the standard single-view-gallery protocol and excludes identical views. The lower panel uses the separately labeled pooled multi-view gallery; "synthetic" averages four corruption families at severities 0.1, 0.3, 0.5, and 0.7.

Standard single-view protocol				
Descriptor	NM	BG	CL	Average
Global only	67.64 ± 2.83	49.50 ± 1.92	27.91 ± 0.94	48.35 ± 1.85
Raw parts, uniform	74.02 ± 3.04	55.56 ± 2.04	32.54 ± 1.88	54.04 ± 2.21
Raw parts, reliability weighted	74.14 ± 3.04	55.47 ± 2.01	32.52 ± 1.81	54.04 ± 2.18

Pooled multi-view controlled endpoints			
Descriptor	Uncorrupted	Synthetic, clean gallery	Synthetic, occluded gallery
Global only	68.06 ± 0.50	48.90 ± 0.63	54.34 ± 0.63
Raw parts, uniform	74.76 ± 0.47	55.13 ± 1.22	59.88 ± 0.90
Raw parts, reliability weighted	74.64 ± 0.45	55.29 ± 1.22	59.96 ± 0.90

Table 6 answers the fifth study question without changing the selected objective, descriptor, or fusion weight. On the standard protocol, reliability-weighted raw parts improve the three-condition average over global matching by 5.69 percentage points; the paired five-seed 95% interval is $[5.19, 6.20]$. The condition-specific gains are 6.50 points for NM, $[6.16, 6.84]$; 5.97 for BG, $[5.73, 6.22]$; and 4.61 for CL, $[3.51, 5.71]$.

The saved test-identity outcomes permit stronger paired analysis for the pooled protocol. Reliability-weighted raw parts gain 6.58 points on uncorrupted

probes, with crossed-bootstrap interval $[5.37, 7.83]$; 6.39 points over the positive-severity synthetic grid with a clean gallery, $[5.40, 7.40]$; and 5.63 points with independently corrupted galleries, $[4.89, 6.40]$. Uniform and reliability- weighted raw matching remain practically tied on the standard average: the reliability-minus-uniform difference is $+0.004$ points with paired interval $[-0.059, +0.067]$. Reliability gives small positive synthetic-grid differences of 0.16 points for the clean gallery and 0.08 points for the occluded gallery, but these do not alter the mechanism conclusion: anatomical score fusion provides the material gain.

4.6 Efficiency

The common backbone contains 1.69 million parameters. Mean training time is 1.63 hours for the metric baseline, 1.89 hours for global SupCon, and 1.97–1.99 hours for the combined or part-triplet variants on the recorded H100 MIG environment. This is approximately 21% overhead for the combined model, substantially below an initial twofold estimate. These measurements do not establish deployment cost: the reliable-parts descriptor stores and scores five additional local embeddings, so inference latency, gallery storage, and pairwise scoring throughput must be measured before an efficiency claim is made.

5 Discussion

5.1 What the factorial establishes

The controlled comparison rejects the original hypothesis that adding reliability-gated part SupCon to the selected metric baseline improves occlusion robustness. The conclusion is stronger than a single unfavorable mean: the degradation is visible across descriptors and gallery protocols, and the paired identity-level interval excludes zero for the principal corrupted- probe aggregate. Global SupCon is comparatively benign but offers no demonstrated advantage. Ungating helps the combined objective but does not restore the baseline, while part triplet reveals a separate collapse mode.

The most constructive outcome is the separation between training and matching. Local pooling plus reliability-weighted scoring can help without an explicit local objective, whereas the tested local objectives degrade the representation. The checkpoint-only controls sharpen this suggestion: the best local evidence is already present before a part head. The raw descriptor changes neither the backbone nor its training objective, adds no learned parameters, and improves all four corruption families. Random nonlinear heads preserve part of this geometry, while the tested supervised heads distort or collapse it.

The official test extends the positive matching result beyond the development identities, but it does not retest all six training objectives. The negative conclusion about auxiliary part training therefore remains a controlled development result; the official confirmation concerns the selected metric baseline and the three matching descriptors.

5.2 Official confirmation and remaining evaluation

The candidate carried into official evaluation is reliability-weighted normalized raw-part pooling with a 0.5 global–local score mixture. Its gain is preserved across five new full-training runs and all three standard CASIA-B conditions. Uniform weighting remains an important ablation because its performance is nearly identical; we do not claim that reliability is a major independent contribution. Likewise, the 0.75 mixture is reported only as development sensitivity and is not promoted after observing validation.

A frozen-backbone head-only study remains useful as a mechanism experiment. Global-to-part stop-gradient distillation and a low-weight part triplet with an explicit variance floor can test whether a learned head can retain the raw geometry. It is no longer a prerequisite for defining the primary method: if neither head exceeds the fixed raw descriptor without violating clean accuracy or the 10^{-4} variance floor, the raw descriptor remains the confirmatory candidate. The official CASIA-B test has now been consumed and should not guide further tuning. The next priorities are real occlusion, an established external backbone, and another dataset rather than another broad loss-weight sweep.

6 Limitations and Validity Threats

Single-dataset official confirmation. The study reports the official CASIA-B test identities for the selected metric baseline and matching descriptors, but not for all six development objectives or reproduced state-of-the-art systems. The official experiment confirms within-dataset transfer of the matching decision; it is not by itself evidence of cross-dataset generalization or a leaderboard claim.

Synthetic rather than real occlusion. The corruptions are controlled missingness mechanisms. They do not reproduce left/right swaps, cross-person joint assignment, smooth but anatomically wrong tracks, or population and camera shifts. At least one real-occlusion endpoint is necessary before claiming operational robustness.

One pose source and one backbone. All runs use the same HRNet-derived 2D pose and an internal multistream graph network. A general claim that part supervision is harmful requires replication with another pose source, dataset, and established backbone. Conversely, a positive redesigned method must outperform reproduced external baselines rather than only this internal control.

Three development seeds and five confirmation seeds. Three seeds support the factorial diagnosis but provide limited precision for model-training variability. The official comparison uses five matched seeds; its paired standard-protocol interval still has only four degrees of freedom. Identity resampling in the pooled-outcome bootstrap represents test-population variation but does not create additional independent model trainings. No family-wise error claim is made over the many exploratory corruptions and descriptors.

Descriptor development on the same validation identities. The raw descriptor, uniform/reliability ablation, random-head controls, and fusion sensitivity are all evaluated on the same 12 development identities. The primary 0.5 fusion weight and acceptance thresholds were declared before running these controls, but choosing the raw descriptor still constitutes model development. The higher validation score at weight 0.75 must not be treated as confirmatory. The fixed 0.5 candidate has now been evaluated on the official test; those outcomes must not be used for another tuning round. External datasets remain required. Gradient cosines are computed at the shared paired-view feature tensor; clean-anchor CE occupies a different batch slice and is therefore orthogonal by construction at that tensor. We consequently do not interpret clean-CE cosine values. Parameter-level diagnostics are needed for any new joint training.

Reproducibility provenance. The factorial records source SHA-256 "ad96b902..." and the descriptor controls record "eddf23a2..."; both use dataset SHA-256 "123e838c...". The official confirmation records source SHA-256 "7a351cac...", dataset SHA-256 "9823d021...", and configuration fingerprint "f25bd174...". Every condition retains its configuration, environment, selected epoch, aggregate metrics, and probe outcomes. The five official runs used two H100 NVL MIG profiles but identical software, source, data, and configuration hashes. However, the runs record "not-a-git-checkout". The public repository should therefore tag the exact source snapshot, lock dependencies, and provide a manifest connecting every reported cell to its run directory.

Code availability. The implementation, experiment configurations, SLURM workflows, and aggregation utilities are publicly available at `https://github.com/likwifi/oap-supcon`. Dataset files are not redistributed and remain subject to the original provider's access terms.

7 Ethical Considerations

Gait is biometric information and can enable remote identification without an individual's awareness. Improvements in occlusion handling do not remove the requirements for lawful purpose, consent where applicable, access control, retention limits, and human oversight. Evaluation should include demographic and disability-related audits, including mobility aids and atypical gait. The present CASIA-B data cannot support fairness or deployment claims, and the reported synthetic robustness should not be interpreted as operational reliability.

8 Conclusion

In a controlled six-objective, three-seed study, the tested part-level supervised contrastive objective does not improve occlusion-robust pose gait recognition. It substantially reduces robustness relative to a shared classification-plus-triplet baseline. Reliability gating is not the sole cause, global SupCon is inconclusive,

and part-triplet training collapses the local embeddings. Checkpoint-only controls resolve the apparent random-head confound: normalized raw part pools improve clean Rank-1 by 2.95 points and the principal corrupted-probe aggregate by 4.32 points at the declared 0.5 fusion weight, with paired intervals excluding zero. Ten fresh random projections also help, but less. Uniform and reliability-weighted raw matching are nearly tied, so anatomical pooling and score fusion—not strong evidence for confidence gating—form the supported mechanism. The justified candidate is therefore a globally trained backbone with parameter-free raw part matching. Five-seed official-test evaluation confirms this candidate: under the standard single-view protocol, reliability-weighted raw parts raise the three-condition average from $48.35 \pm 1.85\%$ to $54.04 \pm 2.18\%$, a paired gain of 5.69 points with interval $[5.19, 6.20]$. Pooled uncorrupted and synthetic test endpoints show gains of 5.63–6.58 points. Uniform raw matching remains effectively tied, reinforcing that anatomical fusion rather than reliability weighting supplies the material benefit. Real occlusion, an external backbone, and additional datasets remain necessary before a broad robustness claim is warranted.

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